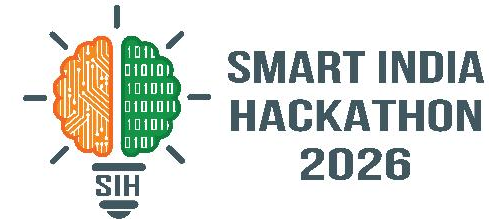
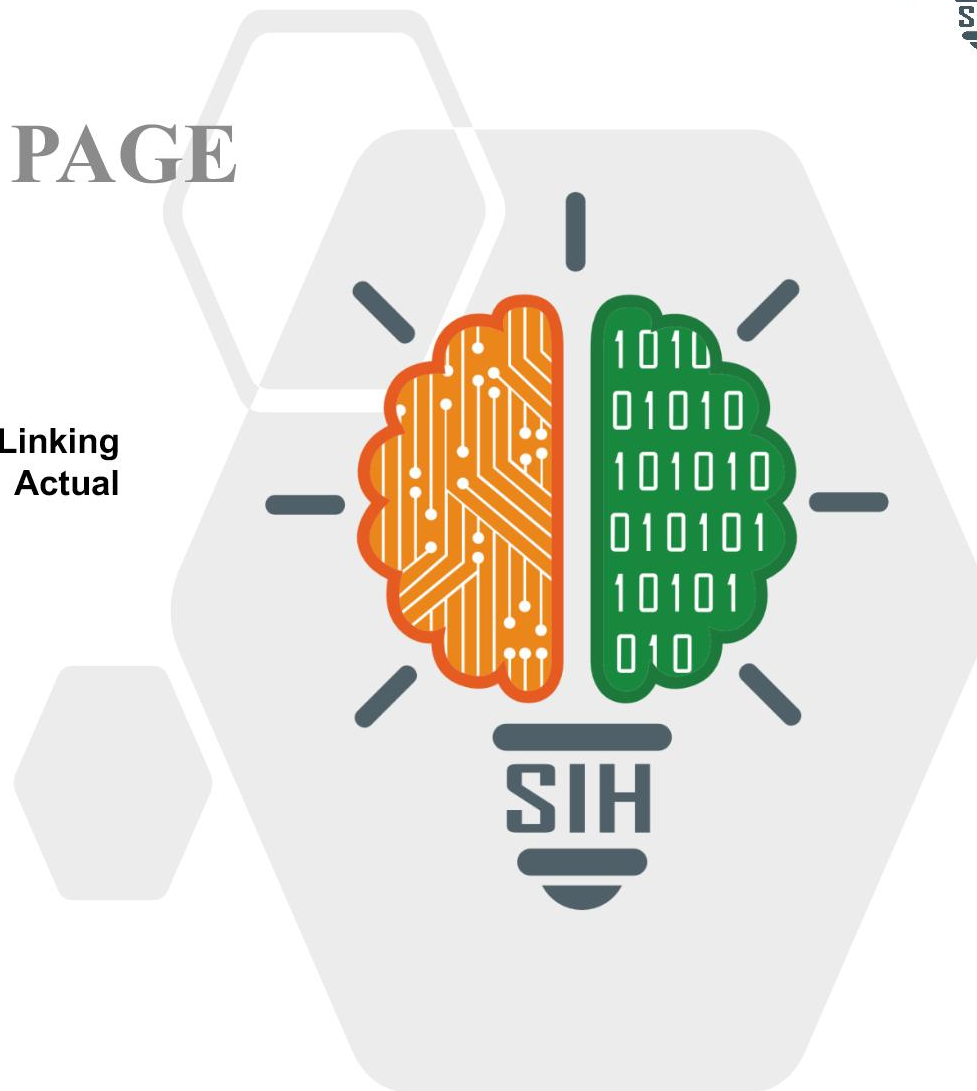


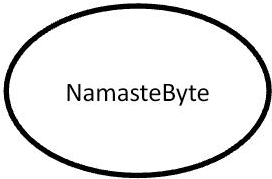
SMART INDIA HACKATHON 2026



TITLE PAGE

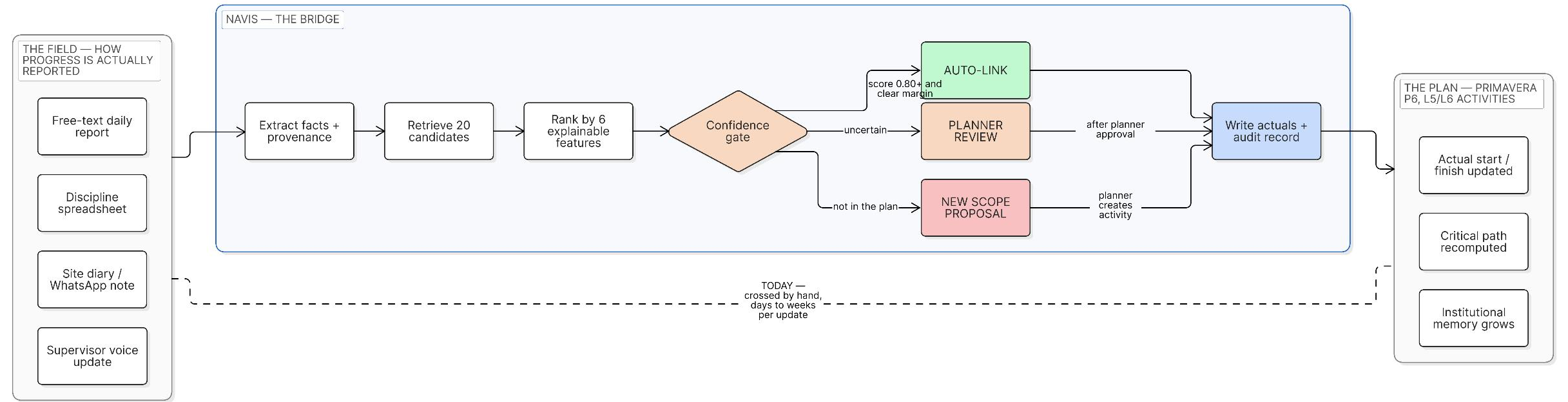
- **Problem Statement ID – 26122**
- **Problem Statement Title- Intelligent Data Capture & Schedule-Linking Layer for Infrastructure Project Management: Real-Time Actual Progress Tracking (Planning-to-Execution Bridge)**
- **Theme- Smart Automation**
- **PS Category- Software**
- **Team ID-**
- **Team Name- NamasteByte**





PROPOSED SOLUTION

NAVIS - one evidence trail from a field sentence to a reviewed P6 actual.



THE PROBLEM

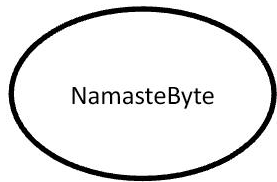
Progress is reported as prose - daily reports, discipline sheets, voice notes. Primavera P6 needs an activity ID and a date. A planner bridges that gap by hand, every cycle, on a schedule of thousands of activities.

HOW NAVIS SOLVES IT

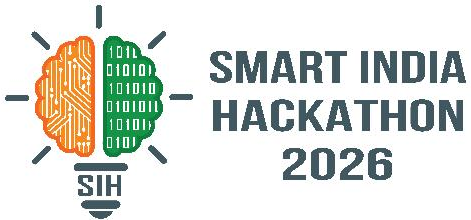
Extract the fact with the file and line it came from, retrieve 20 candidate activities, rank on 6 explainable features, then gate on confidence. Actuals are written with an append-only audit record.

WHAT IS NEW - REFUSAL IS THE FEATURE

100% auto-link precision, 67 of 67 on a held-out test. The rest is queued for a planner, never guessed. A wrong actual date triggers false billing or a premature hydrotest - so the system asks instead.



TECHNICAL APPROACH

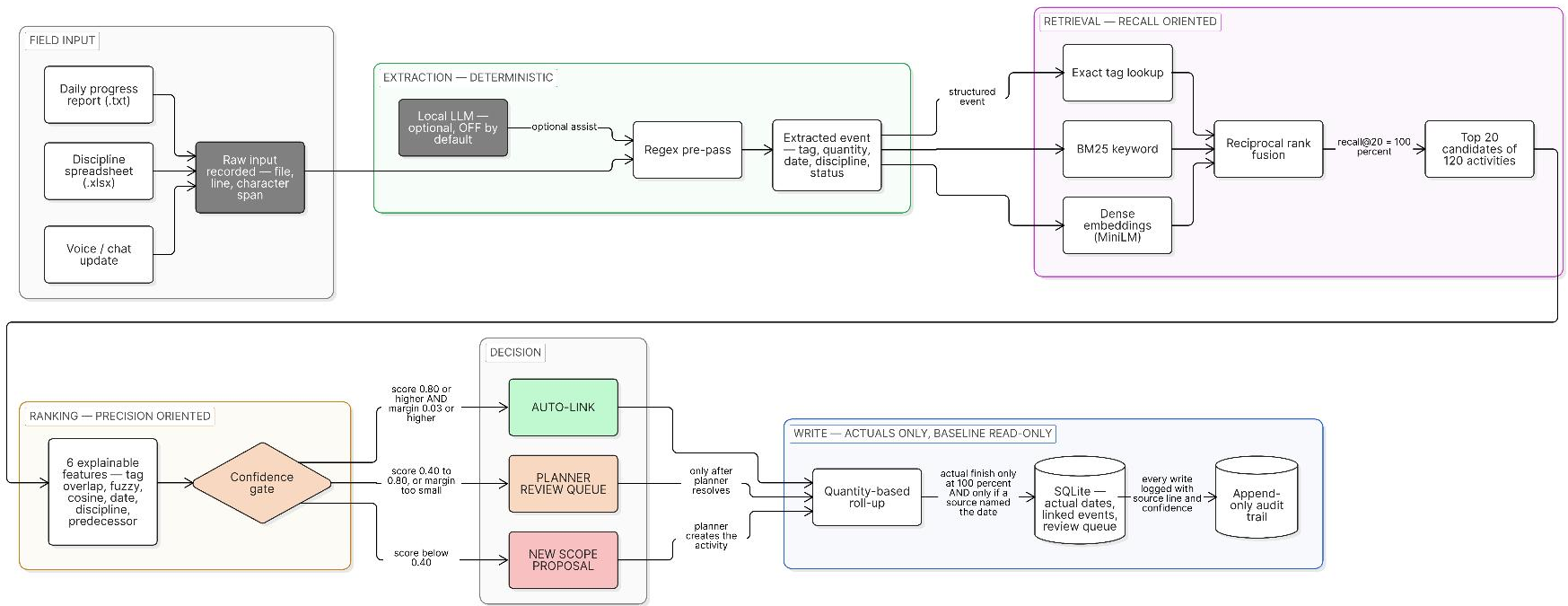


TECHNOLOGIES USED

- React + TypeScript - Vite front end
- FastAPI - 46 HTTP routes, live OpenAPI docs
- Deterministic rules extraction - .txt, .xlsx, .pdf, .csv, image OCR
- Hybrid retrieval - exact tag + BM25 + MiniLM embeddings, RRF fused
- Ranking - 6 explainable features, no black box in the loop
- SQLite - actuals, review queue, append-only audit
- Primavera P6 PMXML and XER - pure Python, no MPXJ, no JVM
- Local LLM via Ollama - optional, OFF by default

Runs on one laptop CPU. No cloud service, no API key.

FLOW CHART



100%

AUTO-LINK PRECISION

67 of 67 correct

43.5%

COVERAGE

67 of 154 mentions

86.9%

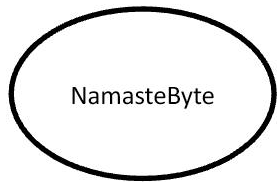
TOP-1 ACCURACY

126 of 145

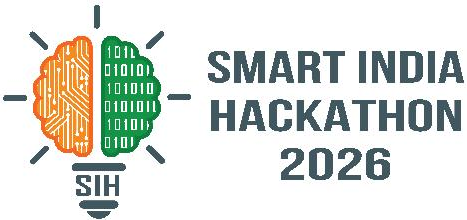
Verify python eval.py reproduces every figure above

Scope 154 held-out mentions, synthetic corpus, one project

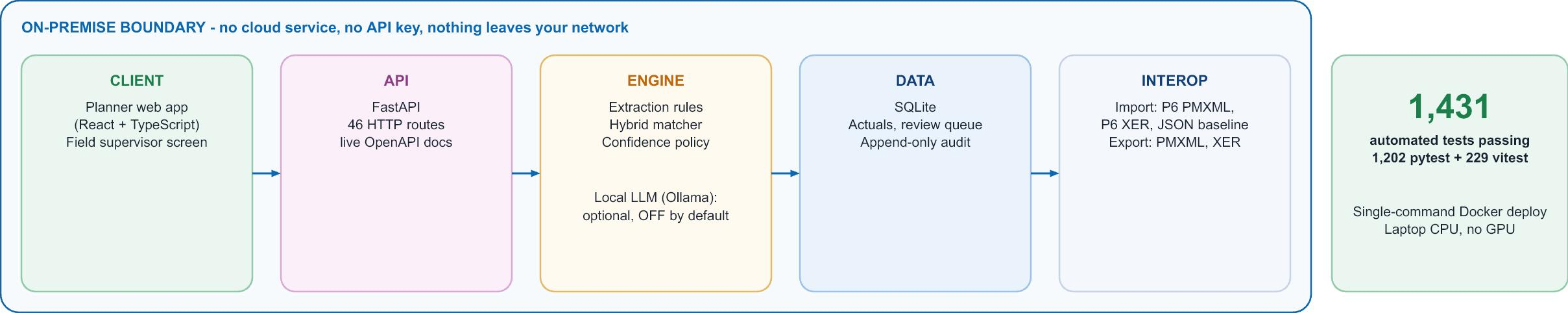
Working prototype - 1,431 automated tests passing



FEASIBILITY AND VIABILITY



FEASIBILITY - IT RUNS ON ONE LAPTOP No cloud service, no API key, nothing leaves your network.



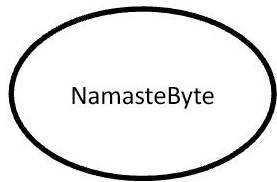
CHALLENGES AND RISKS

- Field wording never matches the WBS text
- A wrong actual date means false billing or a premature hydrotest
- Work happens that nobody planned
- No live Oil India schedule is available to us
- Site or venue has no internet

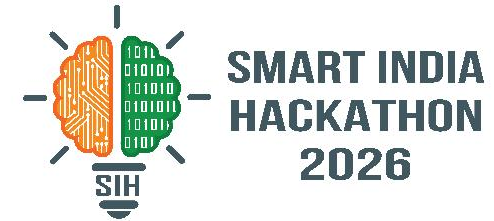
MITIGATION - ALREADY BUILT

- Three retrieval channels fused - the correct activity is inside the top 20, recall@20 = 100%
- Nothing is auto-written unless score, margin and discipline all agree - 0 wrong auto-links
- Raised to the planner as a new-scope proposal, never dropped silently
- Synthetic schedule structured on the public CFIHOS oil and gas taxonomy - and we say so
- The whole stack runs on one laptop CPU - no cloud, no GPU, no API key

Known gaps, stated: no authentication (the login screen is a role picker), no offline queue, no .mpp import, synthetic labels.



IMPACT AND BENEFITS



THE LOOP THAT CLOSSES Today this knowledge dies in a supervisor's notebook at project close.



WHO GAINS, AND WHAT THEY GAIN

SITE SUPERVISOR

Speaks one sentence instead of filling a form. Reporting stops being a second job.

PLANNING ENGINEER

Reviews ranked exceptions instead of matching thousands of rows by hand.

PROJECT MANAGER

Sees a schedule that reflects this week, not last month - slippage while it can still be acted on.

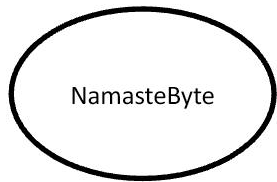
PSU / OWNER

Every actual date is defensible with a file, a line and a confidence.

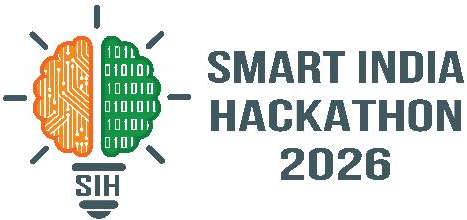
SCALE OF THE EXPOSURE

1,948 monitored central-sector projects, ₹41.99 lakh crore latest revised cost (MoSPI PAIMANA, February 2026).

Context only. NAVIS has run no field pilot, so no saving, ROI or productivity figure is claimed.



RESEARCH AND REFERENCES



REFERENCE AND RESEARCH WORK

MoSPI PAIMANA
Official central-project monitoring portal

used for: the scale figure on slide 5

GAO Schedule Assessment Guide (GAO-16-89G)
What makes a schedule reliable

used for: why actuals must be traceable

NIST AI Risk Management Framework
Traceability and human oversight

used for: why a human confirms every uncertain row

Oracle Primavera P6 EPPM documentation
PMXML and XER interchange formats

used for: our pure-Python parsers

CFIHOS oil and gas taxonomy
Public equipment and discipline taxonomy

used for: the structure of the synthetic schedule

OUR OWN EVIDENCE - WHY COVERAGE IS UNDER HALF

154 held-out test mentions
six source files, no threshold tuned on them

▼

67 auto-linked
43.5% coverage at 100% precision

▼

39 schedule nodes touched
after quantity-based roll-up

32 had no measurable quantity
no percentage, no date written

7 finish dates withheld
the only date came from a report header

Every narrowing step is a deliberate refusal. Reproduce it live: [python eval.py](#)

EVIDENCE BOUNDARY

Synthetic schedule and reports

No live Oil India data yet

In-distribution benchmark

1,431 tests - 1,202 pytest, 229 vitest